SYLLABUS

Course Information

Process Safety Management & Analysis - 2025; CHE 420 / 597; CRN 13096
Instructor: Dr. Ray Mentzer
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Class: TuTh 9:00 – 10:15; FRNY G140

Course Description

The critical importance of Chemical Process Safety is widely recognized after many significant world-wide industrial incidents, such as train derailment and chemical spill in Ohio winter 2023, the massive fertilizer explosion in Beirut, and numerous incidents in the last few years claiming lives, harming the environment, with significant loss of property. All aspects of chemical process safety and loss prevention are addressed in this course. Process safety is concerned with the identification of potential hazards and risks associated with equipment and chemical processing across a variety of industries: oil & gas, chemicals, pharmaceuticals, agriculture, etc. It includes methods for predicting the possible severity of incident scenarios and preventing or mitigating them. The material is thus different from personnel safety, which addresses slips, trips and falls.

As such, it is necessary to understand the operation of these processes and equipment, and to apply sound engineering fundamentals to the analysis and prediction of performance, under adverse circumstances. Thus, the course emphasizes quantitative engineering analysis, and in a broader context critical thinking, complex reasoning and written communication. This is based on the application of mass and energy balances, fluid mechanics of liquid / gas / two-phase flow, heat transfer and the conservation of energy, mass transfer, reaction kinetics, process control, and diffusion & dispersion under highly variable conditions.

Techniques for performing process hazard analysis, risk assessment, and accident investigations are covered, including the review of numerous significant industrial incidents. All students will perform an incident investigation of a significant industrial incident. CHE 597 graduate students have additional homework problems and a research project with a written report.

Learning Resources, Technology & Texts

Course topics follow those in the text: Daniel A. Crowl and Joseph F. Louvar, **'Chemical Process Safety: Fundamentals with Applications'**, Prentice Hall (4th edition), which will be supplemented with other pertinent materials, such as videos of incident investigations by the US Chemical Safety Board. Pertinent class materials will be posted on Brightspace.

Types of Assignments / Assessments

Teams: Students will be grouped into teams of ~3 by the CATME system, and work on homework assignments and project(s) as a team. **Every student will submit homework,** which will **not** be identical among teammates**.** The purpose of working in teams is not to 'spread the work around' but to capture the synergies of teamwork, benefiting from each member's perspective. Team members will periodically evaluate each other in terms of their contribution to homework and team project(s), which will be reflected in course grades.

Homework: Homework will generally be assigned each week and **due at noon ET on Friday's**. Late homework will **not** be accepted, except with prior approval. Homework will be submitted,

graded and solutions posted on Brightspace. Copying from a Solutions Manual or classmates will not be tolerated, with a zero given on homework assignments for infractions.

Attendance: Class attendance is important. Class participation is encouraged, material will be covered in class beyond the text, including several videos and there will be unannounced quizzes. Notify the instructor **in advance** for all excused absences (e.g., job interviews).

Incident Investigation: Each team will investigate a major incident and prepare a report (selected from a list of incidents provided or another of their choice). Topics will be distributed during the 4th week of class, teams will submit their top three choices by September 23, with the final topic confirmed by the Instructor on September 25. The report will be due on October 28. Grading of the report is based on the written report and teammate evaluations of individual contributions.

Project: ***For 597 students only.*** Each team will prepare a research project (selected from a list of topics provided or another of their choice) and submit a formal report. Topics will be distributed the 6th week of class, teams will submit their top three topics by October 9, with the final topic confirmed by the Instructor on October 16. An abstract is due on November 11 and the report due on December 9, both electronically and as a hard copy. Grading of the project is based on the abstract, written report, and teammate evaluations.

Exams & Quizzes: There will be three exams and three quizzes during the term, all given during scheduled class time in G140. **All exams will be open-book and thus it is imperative that each student have access to a hard copy of the text (no xeroxed or on-line copies).** Exam III will not be comprehensive. Missing a quiz / exam results in a zero, unless arrangements were made with the Instructor in advance. Regrade requests for exams / quizzes and homework should be made to the party who did the initial grading, the TAs for the former and Graders for the latter. Any such requests must be made within two weeks of the posting of the graded work product in Brightspace.

Grading Criteria:

	CHE 420	CHE 597
Homework	20%	20%
Incident Investigation	15%	10%
Project Report	0%	10%
Quizzes	10%	5%
Exams	55%	55%

Grades:

90-100	A
80-89	B
70-79	C
60-69	D
<60	F

Grading may also include +/- for each grade level.

Support: Professor Ray Mentzer (rmentzer@purdue.edu)

TAs – Megha Das (das239@purdue.edu)

Christopher Dean (dean122@purdue.edu)
Tyler Downard (tdownard@purdue.edu)
Zaid Imran (imranz@purdue.edu)
Katherine Young (young514@purdue.edu)

Graders – Aum Samir Patel (pate2457@purdue.edu)
Rayan Hussain (hussair@purdue.edu)

The TAs will conduct two one-hour weekly help sessions: **Wed, 11:00 – 12:00 & Thurs, 3:00 – 4:00.**

Academic Integrity

It is tantamount that students reflect on and adhere to the **Purdue Honors Pledge**, *"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."* Integrity / ethics violations such as plagiarism, copying from others work, or use of unauthorized online services (e.g., Chegg) will not be tolerated. Penalties include up to a failing grade and notifying the Dean of Students. While students may use AI tools for research, written material from ChatGPT, etc may not be incorporated within class submissions.

Specifically, Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Section B-2-a, Code of Student Conduct] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972] All incidents of academic dishonesty will be reported to the Dean of Students. Such incidents include: i) possessing or accessing, in hardcopy or electronic form, the solution manual to the course text, or to the exams, ii) claiming credit for work that is not your own original work, and iii) enabling other students to create work that is not their original work.

Non-Discrimination Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach their potential.

Accessibility Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let the instructor know and to discuss options. You are also encouraged to contact the Disability Resource Center at drc@purdue.edu or by phone: 765-494-1247.

Mental Health / Wellness If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell.

Emergency Preparation For a fire alarm evacuate and assemble near the Awards Tower in the Engineering Mall. For a shelter in place, per outside siren, move to the basement of FRNY.

CHE 420 / 597 Chemical Process Safety (Fall, 2025)

Course Schedule

Date		Topics	Note
August	Wk1a-8/26	Why study process safety? Syllabus & Chapter 1	Piper Alpha video
	Wk1b – 8/28	Chapter 1. Introduction – cont'd	
_____	Wk2a-9/2	Chapter 1. Introduction – cont'd	
	Wk2b -9/4	Regulations & Mgmt Systems	Fatal Exposure – DuPont video
	Wk3a-9/9	Regs; Teamwork	
	Wk3b – 9/11	Chapter 2. Toxicology	**Quiz 1**
September	Wk4a-9/16	Incident Investigations	Bhopal disaster; Incident investigation project introduced
	Wk4b – 9/18	Chapter 3. Industrial Hygiene	BP TX City video; teammate evaluations
	Wk5a-9/23	Chapter 4. Source Models – I	Teams select incidents
	Wk5b – 9/25	Ch #4 Source Models II	Instructor confirms projects
	Wk6a-9/30	**Exam I**	
_____	Wk6b – 10/2	Chapter 5. Toxic Releases	MGPI, Mixed Connection, Toxic Result video; *597 Project topics distributed*
	Wk7a-10/7	Toxic Releases	
	Wk7b – 10/9	Chapter 6. Fires & Explosions - I	Imperial Sugar video; *597 Teams select topics*
October	Wk8a-10/16	Chapter 6. Fires & Explosions – II	**Quiz 2;** *Instructor confirms 597 project topics*
	Wk 9a– 10/21	Chapter 7. Designs to Prevent Fires & Explosions	Static Electricity video
	Wk9b-10/23	Chapter 8. Chemical Reactivity	T-2 Incident video
	Wk10a -10/28	Safe Designs & Operations	Teams submit incident investigation reports & team member evaluations
	Wk10b-10/30	Safe Designs & Operations – cont'd	Blast Waves in Danvers video
_____	Wk11a -11/4	**Exam II**	
	Wk11b-11/6	Chapter 9. Intro to Reliefs	'Without Safeguards Pressure Vessels can be Deadly' CSB video
	Wk12a – 11/11	Chapter 10. Relief Sizing	*597 Project abstract due*
November	Wk12b-11/13	Chapter 10. Relief Sizing -cont'd	Univ Laboratory Safety video
	Wk13a- 11/18	Chapter 11. Hazards Identification – I	West Explosion video
	Wk13b-11/20	Chapter 11. Hazards Identification – I I	**Quiz 3**
	Wk14a- 11/25	Chapter 11. Hazards ID – II – cont'd **Thanksgiving Break**	DeRidder Pulp & Paper video
	Wk15a-12/2	Chapter 12. Risk Assessment	
December	Wk15b-12/4	Chapter 12. Risk Assess. – cont'd	ExxonMobil Torrance Refinery video
	Wk16a – 12/9	Emergency Response	Emergency Preparedness video; *597 Project reports due*
	Wk16b–12/11	**Exam III**	

Learning Outcomes

Our graduates will be able to (**bold** are specific to 420):

1	***Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.***
2	***Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.***
3	Communicate effectively with a range of audiences.
4	***Recognize ethical and professional responsibilities in chemical engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.***
5	Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6	Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7	***Acquire and apply new knowledge as needed, using appropriate learning strategies.***

ABET Syllabus

1. **CHE 42000: Process Safety Management and Analysis**

2. **Credits and contact hours:**
   3 credits
   Lecture – 2 days per week at 75 minutes each for 16 weeks

3. **Instructor's or course coordinator's name:** Professor Ray Mentzer

4. **Textbook(s):**
   a. Daniel A. Crowl and Joseph F. Louvar, **'Chemical Process Safety: Fundamentals with Applications'**, Prentice Hall (4th edition)

5. **Specific course information**
   a. **Catalog description:** Develop knowledge of process safety management and analysis in the process industries – including hazard identification, hazard analysis and risk management.
   b. **Prerequisites:** CHE 34800, 37800 (both concurrent)
   c. **Course Status:** Required

6. **Specific goals for the course**
   a. **Specific outcomes of instruction**
   - Demonstrate knowledge and understanding of the elements of process safety management (→45000)
   - Apply the techniques, analytical skills, and modern computational tools necessary for performing process safety calculations in the design of safety equipment (20500, 21100, 34800, 37700, 37800→); (→45000)

- Demonstrate an understanding and appreciation of the need for professional integrity and ethical decision making to promote safety in the workplace (20000, 30000, 40000→); (→45000)
- Be able to pro-actively identify and analyze safety hazards (→45000)
- Demonstrate an understanding of the detrimental effects of the unsafe operation of chemical facilities on environmental, health, and safety issues and other public interests. Our graduates will be aware of the wide-reaching effects that engineering decisions have on society, our global community and our natural environment (20000, 30000, 34800, 37700, 37800, 40000→); (→45000)
- Demonstrate knowledge and understanding of risk management tools, programs and processes associated with process safety. (→45000)

b. Student outcomes addressed by the course:

1) Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.

2) Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

4) Recognize ethical and professional responsibilities in chemical engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

7) Acquire and apply new knowledge as needed, using appropriate learning strategies.

7. Topics

Process Safety Management and Hazard Identification

Regulations – 29 CFR 1910.119

People, Technology, Facilities

Management/Leadership, Required Management Systems

Analysis of Hazards

Numerous Case Studies (14): Piper Alpha, Bhopal, BP TX City, Imperial Sugar, T2 Laboratories, West Explosion, ...

What If, Checklist, Hazard and Operability (HAZOP) Study

Hazard vs. Risk, Qualitative & Quantitative Risk Assessment

Failure Mode and Effects Analysis

Fault Tree Analysis

Risk Management

Safety Culture

Regulations and Audits, Ethics and Safety

Operating Procedures, Safe Work Practices

Asset Integrity

Training, Management of Change

Contractor Management

Operations: Start-up/Shutdown/Unsteady State

Emergency Management

Incident Investigation

Process Safety Calculations and Design

Toxicology and Industrial Hygiene

Liquid & Gas Source Models

Toxic Releases & Dispersion Modeling

Flammability and Explosions & Designs for Prevention

Chemical Reactivity
Sizing Relief Valves
Knockout Drums & Flares

8-27-25